

Predicting Mental Health Treatment-Seeking Behaviour in the Tech Industry

ECS784P — Data Analytics, Coursework 1

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1 Introduction

Mental health disorder represents a growing concern across the global technology workforce. Marsh et al. (2024) demonstrate digital workforce stress factors (e.g., communication overload and technostress) that contribute significantly to poor psychological well-being among technology employees, though a large proportion of affected individuals do not seek professional treatment. Understanding these factors that predict treatment-seeking behaviour, therefore, holds practical significance for employers and policymakers who design workforce mental health interventions.

This project addresses the following research question *"Can workforce environment and demographic factors predict whether a technology industry worker seeks mental health treatment?"*. The hypothesis is that features capturing employer support, differing workplace attitudes, and individualistic demographics can infer a predictive signal to train a binary classifier with accuracy exceeding above the 50% chance baseline. The specific aims are to: (i) preprocess and explore the OSMI 2014 Mental Health in Tech Survey, (ii) train and evaluate a supervised Support Vector Machine (SVM) classifier predicting the binary target variable *treatment*; and (iii) apply K-Means clustering to identify natural respondent groupings without reference to class labels.

2 Literature Review

Razavi et al. (2024) provide a detailed scoping review of machine learning approaches with respect to stress and mental health disorder detection, identifying supervised classification as the dominant model, emphasising that robust preprocessing is critical for survey-based data pipelines. These findings directly motivated the decisions I undertook in this project. Iyortsuun et al. (2023) review methods specifically for mental health diagnosis, identifying Support Vector Machines to be the most frequently applied classifier due to their capabilities of generalisation and resistance to overfitting on structured, high-dimensional data, properties that are well-suited to the 23-feature dataset that I used in this project.

Mitravinda et al. (2022) apply supervised learning ML and risk-based clustering directly to the OSMI Mental Health in Tech Survey, identifying family history and employer-provided resources as the strongest predictors, and demonstrating that clustering can meaningfully extrapolate tech workers by mental health risk profiles - this provides a direct methodological comparison in my project also. Tagoe and Mengistu (2025) compare five supervised ML models on tech workforce mental health prediction, reporting accuracies of 69-81% and confirming employer-provided healthcare and readiness to seek treatment as dominant predictive features. Their benchmark contextualises the classification performance achieved in this project.

3 Data Processing

3.1 Data Source and Description

The dataset is the OSMI 2014 Mental Health in Tech Survey, publicly available on Kaggle. It consists of 1,259 respondents and 27 features covering demographics (age, gender, country), employment type (self-employed, company size, remote work), workplace mental health policies (benefits, anonymity leave provision), and attitudes toward mental health disclosure. The target variable is *treatment*: a binary indicator of whether the respondent has sought professional treatment. Table 1 presents a sample of the raw data before preprocessing.

Table 1: Sample of raw survey data (5 rows, 8 selected features). NaN entries in Self-emp are addressed during preprocessing.

Age	Gender	Self-emp.	Family hist.	Treatment	Work interfere	Employees	Remote
37	Female	NaN	No	Yes	Often	6–25	No
44	M	NaN	No	No	Rarely	>1000	No
32	Male	NaN	No	No	Rarely	6–25	No
31	Male	NaN	Yes	Yes	Often	26–100	No
31	Male	NaN	No	No	Never	100–500	Yes

3.2 Missing Values and Column Removal

Missing values were assessed across all columns; three columns were removed entirely: `Timestamp` (no predictive value), `state` (40.9% missing, redundant with `Country`, which had zero missing entries), and `comments` (87% missing; unstructured free text incompatible with the selected methods). Two categorical columns were filled using the mode, the most appropriate strategy for categorical variables: `self_employed` (18 missing, 1.4%) and `work_interfere` (262 missing, 21%). Following this, no missing values remain. Full before/after missing value counts are visible in the Jupyter notebook.

3.3 Age Cleaning

Inspection of the `Age` column revealed a highly skewed distribution prior to cleaning, giving impossible values including -1726, 329, which are clear clerical entry errors. Respondents outside [18, 80] were removed, retaining 1,251 records. Post-cleaning, ages ranged from 18-72 with a mean of 32.1 and a standard deviation of 7.3. The before and after distributions are shown in the notebook.

3.4 Gender Standardisation

The `Gender` column was a free-text field which produced over 50 inconsistent variants. As a result, responses were standardised via string matching into three categories: Male (990, 79.1%), Female (251, 20.1%), and Other (10, 0.8%), where other captures non-binary and ambiguous responses. The male skew is consistent with technology industry demographics at the time of data collection (Mitravinda et al., 2022) and is hence discussed as a limitation in Section 6.

3.5 Data Exploration Findings

Exploration of the cleaned dataset revealed notable patterns. The target variable `treatment` was near-perfectly balanced (50.5% treatment-seekers vs. 49.5% non-treatment seekers), which is most favourable for a classification problem as it mitigates class imbalance altogether. The respondent age distribution was right-skewed with a median of 31, concentrated between ages 27 and 36. The gender distribution, as mentioned, was heavily male-dominant, reflecting the technology industry's demographic composition of 2014.

3.6 Feature Selection

All 23 remaining features were used for modelling. No features were removed post-encoding as each represents a distinct aspect of the workplace environment, demographic background, or mental health attitude. Removing such features without justification risks discarding potentially useful signals. Feature selection via correlation-based filtering was considered but not applied, as no features showed obvious redundancy after `state` and `comments` columns were removed. This represents a potential aspect where future improvements could've been made.

3.7 Encoding and Normalisation

All remaining categorical string columns were converted to integer codes using `.astype('category').cat.codes`, as required by both SVM and K-Means. All 23 features were subsequently normalised to [0,1] using

`MinMaxScaler`. Normalisation is critical as our models are both distance-sensitive algorithms. Without it, features with larger numerical ranges would dominate the hyperplane and centroid calculations, which would introduce scale-driven bias unrelated to the predictive signal.

4 Learning Methods

4.1 Support Vector Machine (SVM)

Support Vector Classification (SVC) identifies the optimal hyperplane in the feature space that maximises the margin between two classes. The support vectors define the hyperplane position, maximising generalisation to unseen data. The regularisation parameter C controls the bias-variance trade-offs: lower C enforces greater regularisation and therefore a wider margin; higher C reduces regularisation and risks overfitting. A linear kernel was selected, appropriate for this tabular dataset. SVM was chosen because: (i) it is well-suited to binary classification with moderate sample sizes; (ii) its generalisation under regularisation is well-documented for mental health classification tasks (Iyortsuun et al., 2023); and (iii) the balanced class distribution satisfies the assumption that classification accuracy is a meaningful metric.

4.2 K-Means Clustering

K-Means partitions data into K clusters by iteratively minimising within clustered sum of squared Euclidean distances between data points and their assigned centroid. The algorithm follows an Expectation-Maximisation process: the E-step assigns each point to its nearest centroid; the M-step repositions each centroid to the mean of its assigned points, which repeats until convergence. K was determined using the elbow method, plotting inertia as a function of K and selecting the value at which the rate of decrease begins to flatten. K-Means was chosen to complement the supervised analysis, determining whether natural respondent groupings exist without reference to the chosen target label.

5 Analysis, Testing and Results

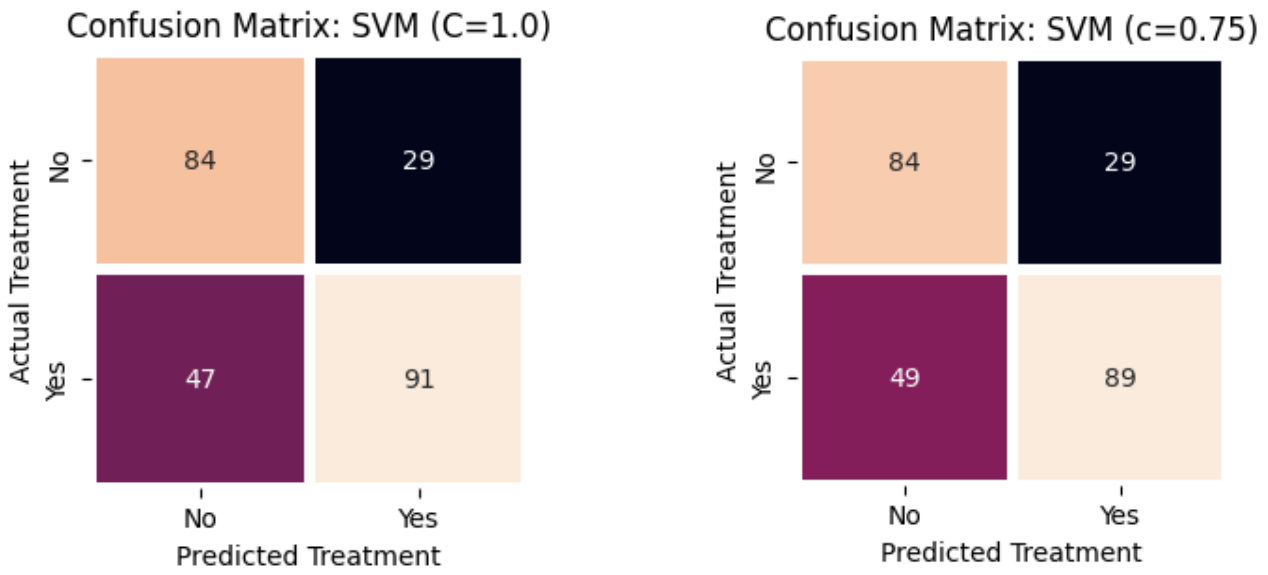
5.1 SVM Evaluation

Data was partitioned into a training set (80%, n = 1001) and held-out test set (20%, n = 250) using `random_state=42` for reproducible results. Five-fold cross-validation was also applied to the training set to obtain a reliable accuracy estimate and to detect overfitting. The baseline SVM (C=1.0) achieved a mean cross-validation of 0.709 (SD=0.027) and training accuracy of 0.722. The small gap confirms no significant overfitting and appropriate model complexity. Table 2 summarises test set metrics for both models.

Table 2: SVM test set performance: baseline (C = 1.0) and regularisation-tuned (C = 0.75) models.

Model	Accuracy	F1-Score	Precision	Recall
SVM (C = 1.0, baseline)	0.697	0.705	0.758	0.659
SVM (C = 0.75, tuned)	0.689	0.695	0.754	0.645

The confusion matrices are shown in Figures 1a and 1b. The baseline model achieved 84 true negatives and 91 true positives, with 29 false positives and 47 false negatives. Note, the higher false negative count infers the model is more prone to predicting non-treatment when treatment is sought. This is directionally asymmetric given the mental health context.



(a) Baseline: C = 1.0

(b) Tuned: C = 0.75

Figure 1: Confusion matrices for SVM baseline and regularisation-tuned models.

To optimize C, a ten-fold cross-validation was performed across $C \in [0.05, 0.95]$ on the training set (Figure 2). Accuracy peaked at *0.75*, however, retraining with this value yielded a marginally lower test accuracy (0.689 vs. 0.697), indicating that C=1.0 was already near-optimal.

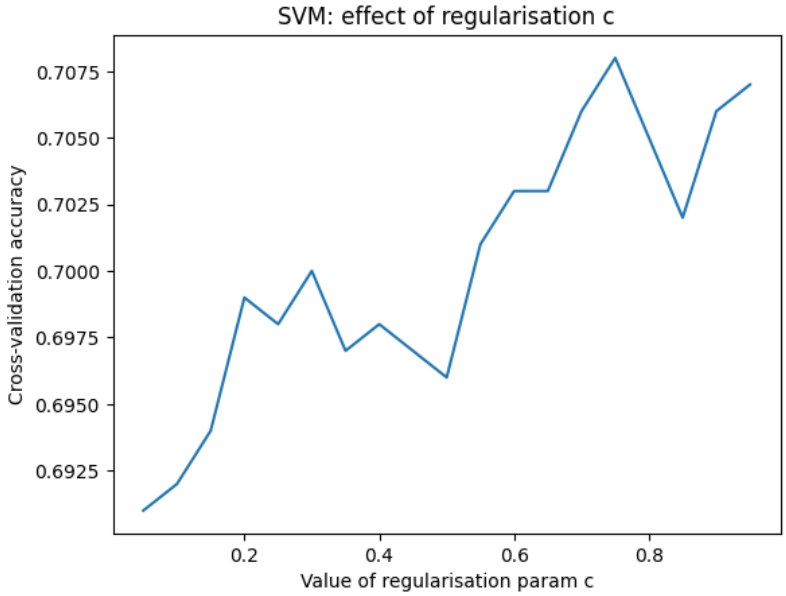


Figure 2: Cross-validation accuracy across regularisation values $C \in [0.05, 0.95]$. Peak at C = 0.75.

5.2 K-Means Clustering

The elbow method was applied across $K \in [1, 10]$ (figure 3). The inertia curve decreased smoothly without an easily identifiable elbow. K=3 was selected where the rate of decrease was visible to flatten. The three clusters contained 553, 396, and 302 respondents respectively, suggesting a large majority group and two smaller subgroups. PCA reduced the 23-dimensional space to two components for visualisation (Figure 4), retaining only 24.6% of total variance. The observed cluster overlap is therefore expected. It reflects the limitations of the 2D projection rather than indicating poor cluster separation

in the original feature space. Both the smooth elbow and the overlapping clusters reinforce the SVM finding that workplace features do not cleanly separate respondents by mental health profile.

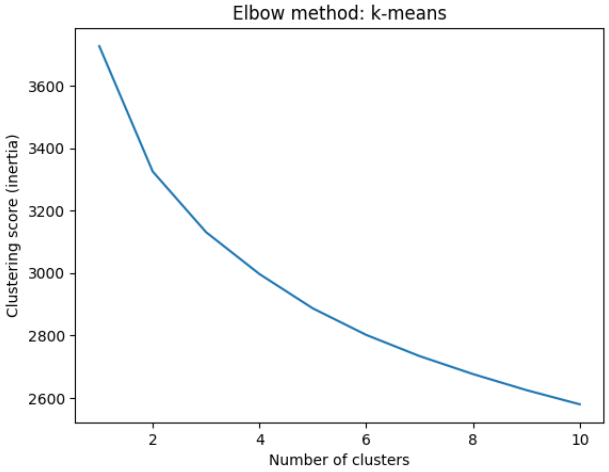


Figure 3: Elbow method: inertia across $K \in [1, 10]$. Smooth curve; $K = 3$ selected.

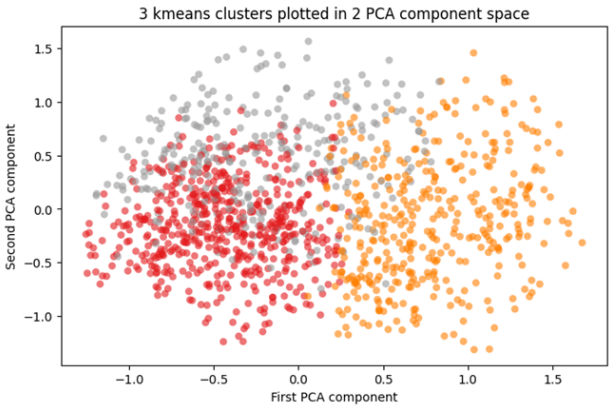


Figure 4: K-Means ($K = 3$) clusters in 2-component PCA space. Only 24.6% variance explained.

6 Concluding Remarks

6.1 Achievements and Challenges

The hypothesis is partially supported: workplace and demographic features can predict treatment-seeking behaviour above chance, but not with the accuracy required for real-world deployment. The SVM classifier achieved 69.7% test accuracy, which exceeds the 50% baseline and falls within the 69-81% benchmark range reported by Tagoe and Mengistu (2025) for comparable tasks. Training and cross-validation accuracies of 72.2% and 70.9%, respectively, confirm good generalisation without overfitting. K-Means identified three respondent groupings, providing unsupervised evidence of partial structure in the data. A key challenge was the quality of the raw data. The Gender column required significant cleaning due to the over 50 free-text responses, and the Age column contained incorrect data entries that had to be removed before any modelling could begin.

6.2 Limitations

Several limitations constrained the results. First, `.cat.codes` encoding assigned integer codes alphabetically to ordinal Likert-scale responses. For example, *Never*, *Often*, *Rarely*, and *Sometimes* received codes 0,1,2,3 which did not preserve the natural ordering and misrepresented the variable structure to the SVM hyperplane. Second, the data are entirely self-reported and subject to social bias, particularly given the stigma associated with mental health disclosure in workplace settings (Marsh et al., 2024). Third, the 79% male, predominantly US-based respondent pool limits demographic generalisability. Finally, only 24.6% of the variance is retained in the PCA projection, excluding the majority of the distribution. Additionally, the dataset chosen was collected in 2014; attitudes toward mental health in the workplace have shifted drastically since, limiting the temporal generalisability of any model trained on this data.

6.3 Future Directions

Future work should employ ordinal encoding to correctly represent Likert-scale ordering. In terms of business applications, an employer could use a trained classifier of this type as part of an anonymous employee survey tool, flagging teams or departments where the predicted proportion of untreated individuals is high, enabling targeted intervention without compromising individual privacy. Incorporating symptom-based clinical data alongside workplace attitudes would provide a richer predictive signal for

such an application. Furthermore, more recent data from the OSMI survey would improve relevance, given that attitudes towards mental health in the workplace have changed considerably.

References

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